

E-001 COVER / LEGEND / DEVICE SCHEDULE

Conv_Simple / CV-101 (garage conveyor) — device schedule · sheet index · wire-numbering key · line-style legend

DEVICE SCHEDULE

Tag	Type	Model	Role	Evidence
PLC1	plc	Allen-Bradley Micro820 2080-LC20-20QBB	Conveyor controller; Modbus RTU master to VFD1	verified
VFD1	vid	AutomationDirect GS10 DURApulse	Drives conveyor motor. Power input is 230 V single-phase (R/L1, S/L2; T/L3 un...	verified
M1	motor		Conveyor drive motor (3~ from VFD1 U/V/W)	field_verify
S0	e stop		Emergency stop, DUAL-CHANNEL (diverse) — NC ch to I-02, NO ch to I-03	verified
SS1	selector switch		Direction selector — FWD contact to I-00, REV contact to I-01 (FWD-OFF-REV)	verified
S2	pushbutton (NO)		Run / rearm pushbutton to I-04; illuminated — lamp fed by PLC O-03 (PBRunLED)	verified
B1	photo eye		Product-present beam to I-05 (blocked -> pe_latched soft stop)	verified
PS1	power supply	Mean Well DIN-rail supply (24 V / 1.0 A; 100-24...	24 VDC control supply (E-004). Feeds the DC +/- distribution block (blue = on...	verified
CB1	circuit breaker		2-pole branch breaker on the 230 V single-phase supply (existence/type/rating...	field_verify
Q1	control relay	Schneider Electric TeSys CA3KN22BD (control rel...	Control relay (Schneider CA3KN22BD, hand-labeled 'MLC') USED AS THE VFD SUPPL...	verified
PL1	pilot light		Green RUNNING pilot — load of O-00 (LightGreen)	verified
PL2	pilot light		Red FAULT/E-STOP pilot — load of O-01 (LightRed)	verified
X1	terminal block		Field terminal strip — field devices land here before PLC (E-008)	field_verify

SHEET INDEX

Id	Title	Status
E-001	Cover / legend / device schedule	drafted
E-002	Power one-line	stub
E-003	VFD power (CB1 -> VFD1 -> M1)	drafted
E-004	24 VDC control power distribution (PS1)	stub
E-005	PLC digital inputs	drafted
E-006	PLC outputs (O-00..O-06)	drafted
E-007	RS-485 Modbus (PLC1 Ch2 <-> VFD1 RJ45)	drafted
E-008	Terminal strip (X1) + wire list	stub
E-009	Open items / field verification	stub

WIRE-NUMBERING KEY

Conductor numbers are W[sheet-digit][line-2d] (e.g. W503 = sheet E-005, line 3) — all PROPOSED until as-built (OI-03/OI-19). Rail/bus labels (W24, W0V) are exempt. Comms sheets (E-007) use mnemonic conductor labels (485+/485-/SGND/SH) per CommsToVFD ancestry. Same electrical node = same wire number; a number changes only through a device.

LINE-STYLE LEGEND

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxxx

proposed wire number (red = unverified)

SAFETY

- LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
- Monitored e-stop is NOT a safety stop — hardwire S0 to remove drive power per NFPA 79 / EN 60204-1 (style law §7; docs/references/industrial-wiring-diagram-standards.md).

COVER / LEGEND / DEVICE SCHEDULE

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM
cover generated from model/*.yaml

SHEET

E-001

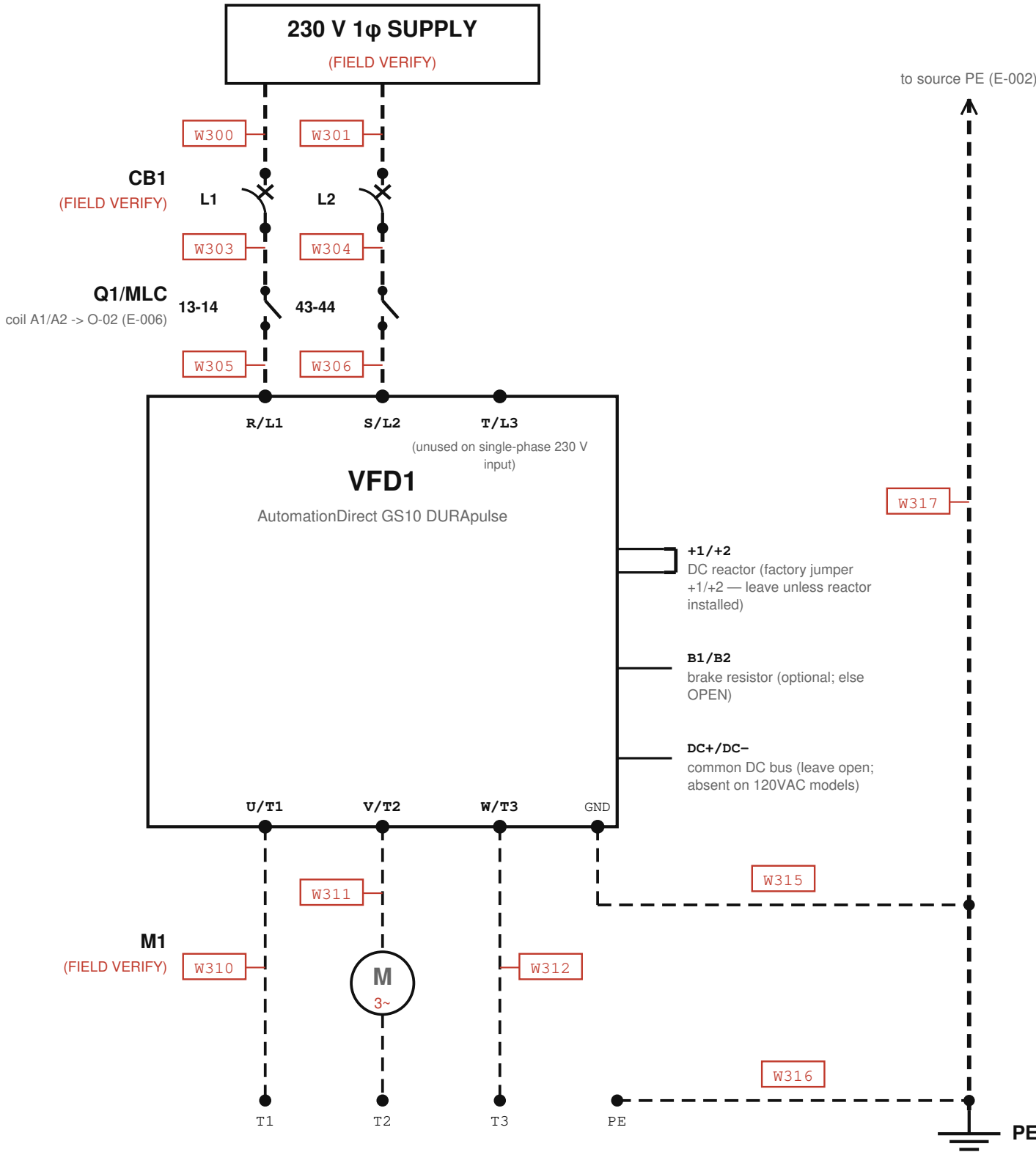
REV A

2026-07-11

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E-003 VFD POWER

Conv_Simple / CV-101 (garage conveyor) — supply → CB1 → Q1/MLC (supply switch) → VFD1 (GS10) → M1 · 230 V 1φ in → 230 V 3φ to motor · every conductor FIELD VERIFY



Bench supply voltage & phase count, GS10 exact model/frame, breaker rating, wire gauge: NOT DOCUMENTED — every conductor FIELD VERIFY. P00.01 = 1.60 A (2026-05-20 export) is the only sizing clue. If 1φ model, input = R/L1, S/L2 only (GS10_UM L1971). MLC (Q1) is a control relay USED AS the single-phase drive-supply switch — its 2 NO contacts (13-14, 43-44) carry the 230 V 1φ supply into the GS10; coil A1/A2 from O-02 (E-006). Energizing O-02 powers the drive (technician-confirmed 2026-07-11).

SAFETY

- Never start/stop via input power (L1811-1813); CB1 is branch protection only, not a run/stop device.
- LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
- Monitored e-stop is NOT a safety stop — hardware S0 to remove drive power per NFPA 79 / EN 60204-1.
- PE resistance ≤0.1Ω (L1760-61); shield/conduit bonded both ends (L1792-95); RFI jumper in/out FIELD VERIFY (L1693-1718, OI-17).

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- proposed wire number (red = unverified)

CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W300	SUPPLY (E-002)	CB1.1	L1 230V 1ph supply	power_line	field_verify	
W301	SUPPLY (E-002)	CB1.3	L2 230V 1ph supply	power_line	field_verify	
W303	CB1.2	Q1.13	L1 protected -> MLC	power_line	field_verify	
W304	CB1.4	Q1.43	L2 protected -> MLC	power_line	field_verify	
W305	Q1.14	VFD1.R/L1	L1 switched -> drive	power_line	field_verify	
W306	Q1.44	VFD1.S/L2	L2 switched -> drive	power_line	field_verify	
W310	VFD1.U/T1	M1.T1	motor phase U	motor_lead	field_verify	
W311	VFD1.V/T2	M1.T2	motor phase V	motor_lead	field_verify	
W312	VFD1.W/T3	M1.T3	motor phase W	motor_lead	field_verify	
W315	VFD1.GND	PE bus	drive PE (≤0.1Ω)	ground	field_verify	
W316	M1.PE	PE bus	motor frame PE	ground	field_verify	
W317	PE bus	SUPPLY (E-002)	PE to source	ground	field_verify	

NOTES

- CB1 REQUIRED per GS10_UM L1758-1759 — type/rating unknown (OI-15).
- M1: swap any two motor leads to reverse — GS10_UM L1773-1776.
- VFD control source = Modbus — see E-007. No control wiring on this sheet.
- Supply is 230 V SINGLE-PHASE (2-wire); drive output = 230 V 3φ to a 230 V motor (technician-confirmed 2026-07-11).

SOURCES:

plc/GS10_UM.txt (terminals L1971-1986; topology L1750-1813) ·
plc/GS10_Integration_Guide.md (Q1 phase-5 test)
plc/Prog_init_ConvSimple_v2.1.st (vfd_run_permit) · plc/MIRA_PLC_WorkInstruction_v3.pdf (LOTO §2)
photos review/photos/wire_1.4.jpg (2026-07-11, actual devices)
review/PHOTO_EVIDENCE_V6.md (technician field correction 2026-07-11)

VFD POWER

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM

terminals per GS10 UM 1st Ed Rev B;

SHEET

E-003

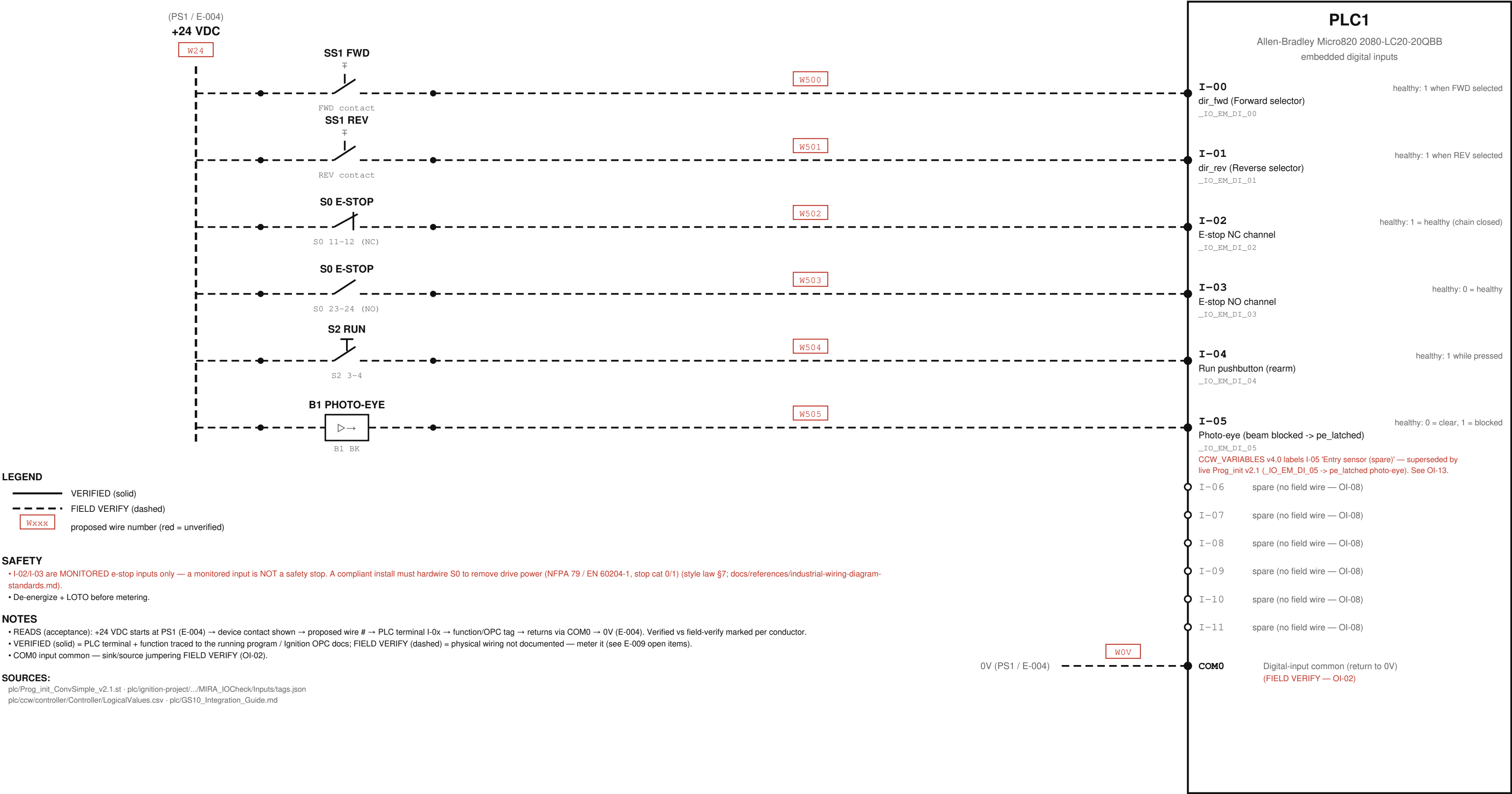
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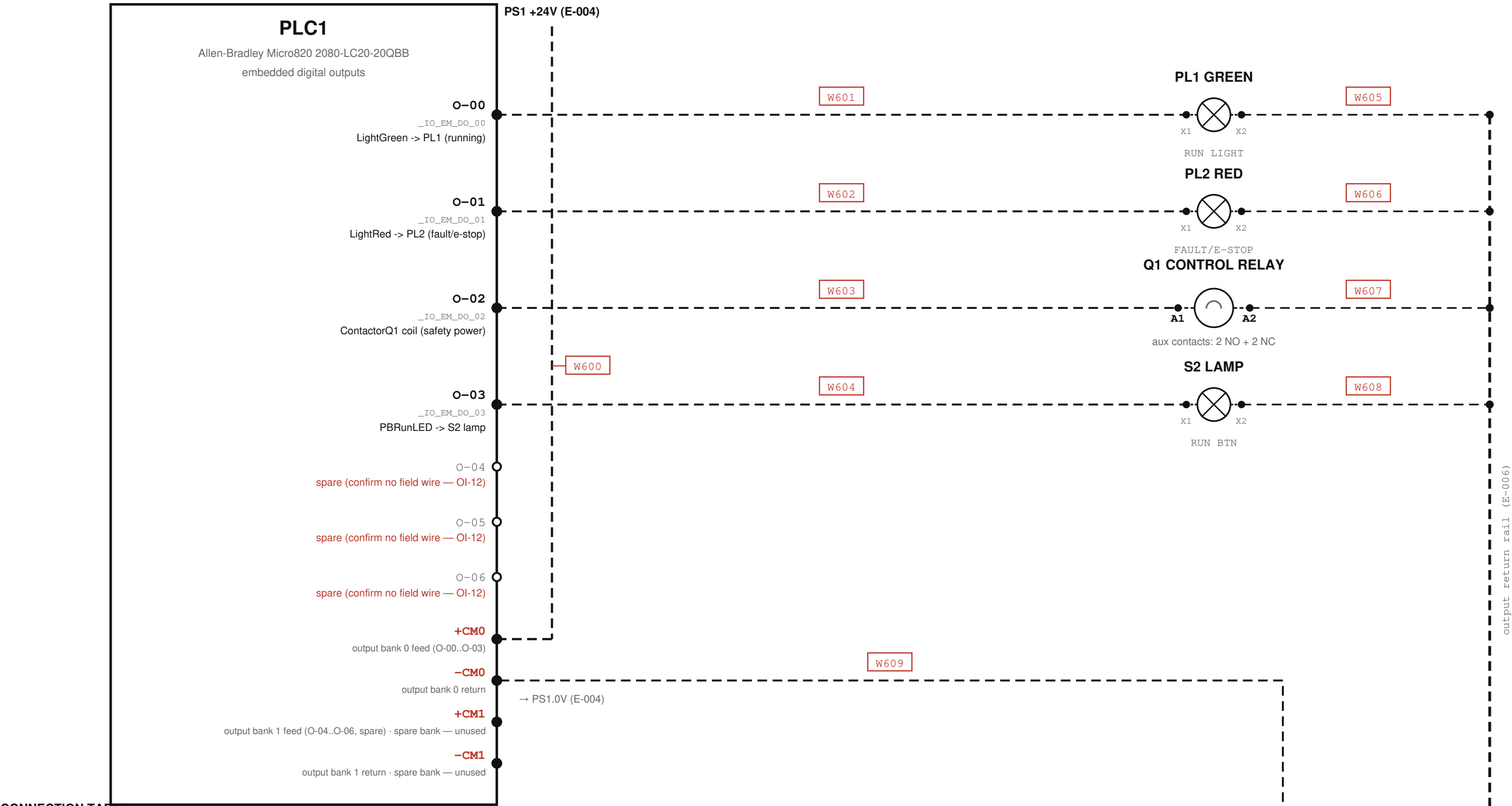
E-005 PLC DIGITAL INPUTS

Conv_Simple / CV-101 (garage conveyor) — Allen-Bradley Micro820 2080-LC20-20QBB embedded DI (24 VDC)



E-006 PLC OUTPUTS

Conv_Simple / CV-101 (garage conveyor) — O-00..O-06 · pilots, Q1/MLC control-relay coil, button lamp · GS10 control = HYBRID Modbus + hardwired I/O (E-007; map OI-22)



Hardwired fallback documented in MIRA-WI-001 v3 §3 Fig 1 (DO_03..DO_07 → GS10 DI1..DI5 + DCM) — NOT ACTIVE and NOT drawn: P02.0x at factory default per the 2026-05-20 parameter export; DO_07 does not exist on the 2080-LC20-20QBB (O-00..O-06); DO_03 collides with PBRunLED in the live I/O map. Bench presence of any such wiring = FIELD VERIFY (OI-18).

SAFETY

- Monitored outputs are NOT a safety function.
- E-stop must remove power per NFPA 79.
- Q1/MLC is a CONTROL RELAY (24 VDC coil, driven by O-02) whose 2 NO contacts (13-14, 43-44) switch the drive's 230 V 1φ supply (V6 correction, E-003) — de-energizing O-02 DOES remove Q1's switched supply to the GS10. Q1 is NOT an NFPA-79/EN-60204-1 safety-rated disconnect (small control-relay contacts, not a lockable safety contactor); the e-stop (S0) → O-02 relationship is FIELD VERIFY (OI-26). Do not treat this circuit as the LOTO isolation point — always LOTO at CB1 or upstream.

NOTES

- Q1/MLC = Schneider TeSys CA3KN22BD control relay (2 NO + 2 NC, 24 VDC coil); coil A1(+)/A2(-) driven by O-02 (verified from photo_wire_3). NO contacts 13-14/43-44 switch the VFD's 230 V 1φ supply (see E-003); 21-22/31-32 NC unused.
- GS10 control (full-res photo): run/dir/speed = Modbus — top FWD/REV/DI connector is EMPTY, RJ45 landed. Bottom analog/output row (+10V/ACM/AI/AO1/DO1/DOC) is wired — likely speed reference + feedback. Map = OI-22.
- NO contacts 13-14 & 43-44 switch the 230 V 1φ VFD supply (see E-003); coil A1/A2 from O-02. NC 21-22/31-32 unused.
- GS10 run/dir/freq = Modbus — see E-007 (P00.20=1, P00.21=2 per 2026-05-20 parameter export)
- No as-built: O-00/O-01/O-03 from CCW v4.0; O-02 corroborated by live Prog_init v2.1.
- Output bank technology + common feed = FIELD VERIFY (OI-09); WI-001 p.4 says relay dry contacts; ±CM polarity naming suggests DC-fed banks — conflicting evidence, resolve per 2080-IN009 + meter.

SOURCES:

- CCW_VARIABLES_v4.0.txt
- Prog_init_ConvSimple_v2.1.st
- MIRA_PLC_WorkInstruction_v3.pdf
- GS10_actual_parameters_5.20.26.xlsx
- GS10_UM.txt
- photos review/photos/wire_1..4.jpg (2026-07-11, actual devices)
- full-res photos review/photos/"_full.jpg + crops (2026-07-11); review/PHOTO_EVIDENCE_V5.md

LEGEND

- VERIFIED (solid)
- - - FIELD VERIFY (dashed)
- Wxxx proposed wire number (red = unverified)

CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W600	PS1.+24V	PLC1.+CM0	output bank 0 feed	power_24vdc	field_verify	
W601	PLC1.O-00	PL1.X1	LightGreen (run light)	control_output	field_verify	
W602	PLC1.O-01	PL2.X1	LightRed (fault/e-stop light)	control_output	field_verify	
W603	PLC1.O-02	Q1.A1	ContactorQ1 coil (safety power)	control_output	field_verify	
W604	PLC1.O-03	S2.X1	PBRunLED (run button lamp)	control_output	field_verify	
W605	PL1.X2	output return rail (E-006)	run light return	control_return	field_verify	
W606	PL2.X2	output return rail (E-006)	fault light return	control_return	field_verify	
W607	Q1.A2	output return rail (E-006)	contactor coil return	control_return	field_verify	
W608	S2.X2	output return rail (E-006)	PB lamp return	control_return	field_verify	
W609	PLC1.-CM0	PS1.0V	output bank 0 return	power_0v	field_verify	

PLC OUTPUTS

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM
output map CCW v4.0 + live Prog_init; O-02 do-not-reuse (WI-001 p.4)

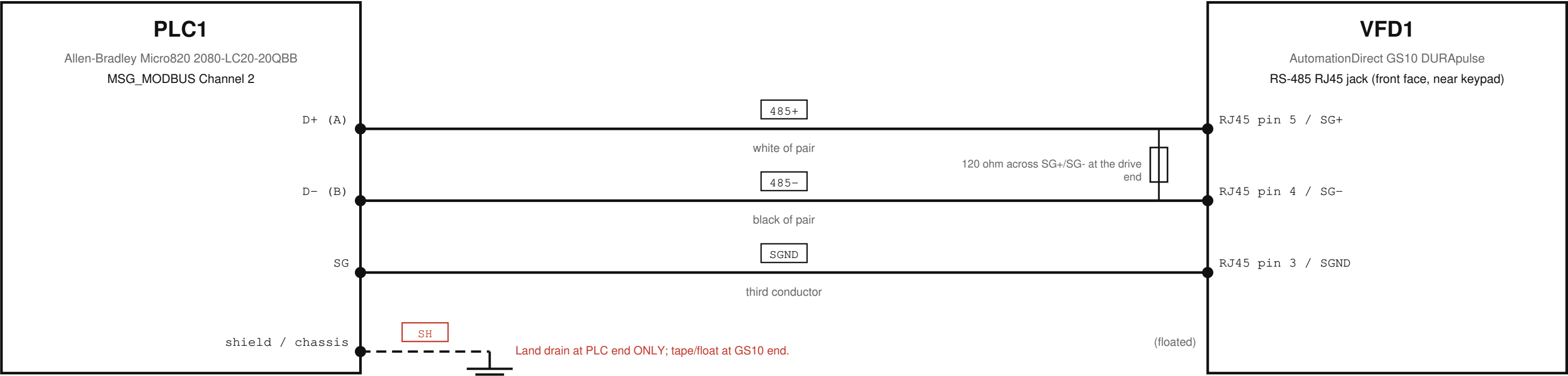
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E-007 RS-485 / MODBUS RTU COMMUNICATION

Conv_Simple / CV-101 (garage conveyor) — Allen-Bradley Micro820 2080-LC20-20QBB (master) ↔ AutomationDirect GS10 DURApulse (node 1) · Modbus RTU only



CONNECTION TABLE

Src dev	Src terminal	Dst dev	Dst terminal/pin	Cable / conductor	Wire	Evidence	Notes
PLC1	D+ (A)	VFD1	RJ45 pin 5 / SG+	Belden 3105A pair · white of pair	485+	verified	Polarity: if silent/garbled, swap 485+/485- first.
PLC1	D- (B)	VFD1	RJ45 pin 4 / SG-	Belden 3105A pair · black of pair	485-	verified	
PLC1	SG	VFD1	RJ45 pin 3 / SGND	3rd conductor · third conductor	SGND	verified	Dedicated wire, NOT the shield.
PLC1	shield / chassis	VFD1	(floated)	tinned drain · drain	SH	field_verify	Land drain at PLC end ONLY; tape/float at GS10 end.

CCW SERIAL PORT: Modbus RTU, Master · 9600 · 8N1 (P09.04 = 12) · Channel 2 · Node 1 (P09.00 = 1)

Older GS10_Integration_Guide.md said 8N2 (P09.04=13) - SUPERSEDED by bench-verified 8N1.

Line params pending re-verify (OI-20): 2026-05-20 export read 38.4k/8N2; 2026-05-26 bench sniff read 9600/8N1 (shown). Fresh keypad/readback to adjudicate.

COMMAND WORDS (0x2000): STOP = 1 · FWD+RUN = 18 · REV+RUN = 34 · freq register 0x2001

34 NOT 20 — GS10_Integration_Guide.md (2026-03-16) and ControlsToVFD/original PhaseB.st print 20; corrected by PhaseB_V1.4 'FIX 2' (bit 5 REV + bit 1 Run), Beginner_Verify_V2 ('NOT 20'), WI-001 Table 9, and live Prog_init v2.1 (vfd_cmd_word := 34).

Beginner_Verify_V2 §4.4 (GS10 manual p.6); WI-001 Tables 8-10; PhaseB_V1.4.st; Prog_init_ConvSimple_v2.1.st

CORRECTED from May-16 draft: Channel 0 → 2 · SGND pin 1/8 → pin 3 · 8N2 → 8N1. Do not copy the old values.

NOTES

- Belden 3105A — shielded twisted pair (485+/485-) + 1 conductor (SGND) + drain
- VERIFIED (solid) = recovered CommsToVFD + Beginner_Verify p48 + Prog_init v2.1; FIELD VERIFY (dashed) = exact PLC chassis ground point; cable P/N if not Belden 3105A.
- Port locations per CommsToVFD (PLC top-edge connector; GS10 front-face RJ45) — WI-001 says bottom-front. Wayfinding only; confirm on unit (FIELD VERIFY).
- GS10 RS-485 RJ45 cable confirmed landed (photo wire_4). The GS10 control connector (FWD/REV/DI/AI/AO1/DO1) is also wired — see OI-22.
- GS10 RS-485 RJ45 confirmed landed (full-res photo). The GS10 top run/DI control connector is EMPTY — no hardwired run/dir; command is Modbus. Bottom analog/output row is wired (OI-22).

SOURCES:

full-res photos review/photos/*_full.jpg + crops (2026-07-11); review/PHOTO_EVIDENCE_V5.md

DOC REFS:

CCW/MIRA_PLC/docs/instructions/Conv_Simple_CommsToVFD.pdf §2 (recovered style + wiring)
CCW/MIRA_PLC/docs/instructions/Conv_Simple_GS10_Beginner_Verify_V2.pdf p48 (corrected pins/channel)
plc/Prog_init_ConvSimple_v2.1.st (Channel 2, node 1, 0x2103)
plc/GS10_Integration_Guide.md §3 (RJ45 pinout, register map)

TROUBLESHOOTING (this circuit)

- Nothing after config -> swap 485+/485- (A/B polarity; vendors disagree).
- CRC errors -> usually polarity. Complete silence -> usually baud mismatch (confirm 9600 8N1).
- GS10 CE10 / F30 comm-timeout -> no Modbus packets; check wiring, baud, P09.03 timeout.
- 120 ohm termination across SG+/SG- at the GS10 end for long runs (bench <2 m usually fine).
- SGND is a dedicated conductor, not the shield; missing common -> intermittent corruption.

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- proposed wire number (red = unverified)

RS-485 / MODBUS RTU Conv_Simple / CV-101 (garage conveyor)	SHEET E-007
Drawn: MIRA / FactoryLM recovers MIRA-WI-001 / Conv_Simple_CommsToVFD §2	REV A 2026-07-117 of 9